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1. 강의참고

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5.

① $x^2y=1 \Leftrightarrow y=\frac{1}{x^2}$: 우함수

② $x+y=1 \Leftrightarrow y=1-x$: 우함수도 기함수도 아니다.

③ 원점 대칭을 확인하기 위해 $(x+y)^3=xy^2$ 에 $(-x,-y)$ 를 대입하면
 $(x+y)^3=xy^2 \Rightarrow (-x-y)^3=-xy^2 \therefore (x+y)^3=xy^2$
 따라서 원점 대칭인 곡선이다.

④ $x^3+y^3=1 \Leftrightarrow y=\sqrt[3]{1-x^3}$: 우함수도 기함수도 아니다.

6. $\sin\left(2\sin^{-1}\left(-\frac{1}{3}\right)\right)=\sin\left(-2\sin^{-1}\left(\frac{1}{3}\right)\right)$
 $=-\sin\left(2\sin^{-1}\left(\frac{1}{3}\right)\right)$
 $=-\sin(2\alpha)$
 $=-2\sin\alpha\cos\alpha \quad \left(\sin^{-1}\alpha=\frac{1}{3} \Leftrightarrow \sin\alpha=\frac{1}{3}\right)$
 $=-\frac{4}{9}\sqrt{2}$

7. $\sin^{-1}\left(\frac{12}{13}\right)=\theta_1 \Leftrightarrow \sin(\theta_1)=\frac{12}{13}$.

$\cos^{-1}\left(-\frac{5}{13}\right)=\theta_2 \Leftrightarrow \cos(\theta_2)=-\frac{5}{13}$

$\Rightarrow \sin(\theta_1+\theta_2)=\sin\theta_1\cos\theta_2+\cos\theta_1\sin\theta_2$
 $=\frac{12}{13}\times\left(-\frac{5}{13}\right)+\frac{5}{13}\times\frac{12}{13}$
 $=0$

따라서 $\theta_1+\theta_2=\pi$ 이다. $\left(0<\theta_1<\frac{\pi}{2}, \frac{\pi}{2}<\theta_2<\pi\right)$

8. $f\left(\cos\left(\frac{4}{3}\pi\right)\right)=\left(\cos\frac{4}{3}\pi-\sqrt{1-\cos^2\left(\frac{4}{3}\pi\right)}\right)\cos^{-1}\left(\cos\left(\frac{4}{3}\pi\right)\right)$
 $=\left(-\frac{1}{2}-\sqrt{1-\frac{1}{4}}\right)\cos^{-1}\left(-\frac{1}{2}\right)$
 $=-\frac{1+\sqrt{3}}{2}\times\frac{2}{3}\pi=-\frac{1+\sqrt{3}}{3}\pi$

참고. $\cos^{-1}\left(\cos\left(\frac{4}{3}\pi\right)\right)\neq\frac{4}{3}\pi$

9. 모든 실수 x 에 대하여 $\tan^{-1}x+\cot^{-1}x=\frac{\pi}{2}$ 이므로

$\tan^{-1}(\tan 1)+\cot^{-1}(\tan 1)=\frac{\pi}{2}$ 가 성립하여

$\cot^{-1}(\tan 1)=\frac{\pi}{2}-1$ 이다.

10. $\tan^{-1}x=\alpha$ 라 하면 $x=\tan\alpha$ 이고 $\cos\alpha=\frac{1}{\sqrt{1+x^2}}$ 이다.

$1-2\cos^2(\tan^{-1}x)=0 \Leftrightarrow 1-2\frac{1}{1+x^2}=0$

$\Leftrightarrow x^2-1=0 \therefore x=\pm 1$

따라서 x 값의 합은 0이다.

11.

① $-\frac{\pi}{2}\leq x\leq\frac{\pi}{2}$ 에 대하여 $\sin^{-1}(\sin x)=x$ 가 성립하므로

$-1\leq x\leq 1$ 에 대하여 $\sin^{-1}(\sin x)=x$ 가 성립한다. (참)

② $0\leq x\leq\pi$ 에 대하여 $\cos^{-1}(\cos x)=x$ 가 성립하므로

$-1\leq x\leq 1$ 에 대하여 $\cos^{-1}(\cos x)=x$ 가 성립하지 않는다. (거짓)

(반례) $x=-\frac{\pi}{6}$ 일 때, $\cos^{-1}\left(\cos\left(-\frac{\pi}{6}\right)\right)=\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)=\frac{\pi}{6}$ 이므로

$\cos^{-1}(\cos x)\neq x$ 이다.

③ $-1\leq x\leq 1$ 을 만족하는 모든 실수 x 에 대하여

$\sin(\sin^{-1}x)=x$ 가 성립한다. (참)

④ $-1\leq x\leq 1$ 을 만족하는 모든 실수 x 에 대하여

$\cos(\cos^{-1}x)=x$ 가 성립한다. (참)

⑤ $-1< x < 1$ 에 대하여 $\sin^{-1}x=\alpha$ 라고 치환하면 $\sin\alpha=x$ 이므로

$\tan\alpha=\frac{x}{\sqrt{1-x^2}}$ 이 성립한다. (참)

12.

$\sinh^{-1}1=\alpha \Leftrightarrow \sinh\alpha=1$

$\cosh^2\alpha-\sinh^2\alpha=1$ 이 성립하므로 $\cosh^2\alpha=2 \Leftrightarrow \cosh\alpha=\sqrt{2}$ 이다.

13.

$\sum_{n=2}^{\infty}\ln\left(1-\frac{1}{n^2}\right)$

$=\lim_{k\rightarrow\infty}\sum_{n=2}^k\ln\left(\left(1-\frac{1}{n}\right)\left(1+\frac{1}{n}\right)\right)$

$=\lim_{k\rightarrow\infty}\sum_{n=2}^k\left(\ln\left(1-\frac{1}{n}\right)+\ln\left(1+\frac{1}{n}\right)\right)$

$=\lim_{k\rightarrow\infty}\ln\frac{1}{2}+\ln\frac{3}{2}+\ln\frac{2}{3}+\ln\frac{4}{3}+\dots+\ln\left(1-\frac{1}{k}\right)+\ln\left(1+\frac{1}{k}\right)$

$=\lim_{k\rightarrow\infty}\ln\frac{1}{2}+\ln\left(1+\frac{1}{k}\right)$

$=\ln\frac{1}{2}=-\ln 2$

14.

$\sum_{n=1}^{\infty}[\tan^{-1}(n+1)-\tan^{-1}(n-1)]$

$=\lim_{k\rightarrow\infty}\sum_{n=1}^k[\tan^{-1}(n+1)-\tan^{-1}(n-1)]$

$=\lim_{k\rightarrow\infty}\tan^{-1}2-\tan^{-1}0+\tan^{-1}3-\tan^{-1}1+\tan^{-1}4-\tan^{-1}2$

$+\dots+\tan^{-1}k-\tan^{-1}(k-2)+\tan^{-1}(k+1)-\tan^{-1}(k-2)$

$=\lim_{k\rightarrow\infty}-\tan^{-1}0-\tan^{-1}1+\tan^{-1}k+\tan^{-1}(k+1)$

$=0-\frac{\pi}{4}+\frac{\pi}{2}+\frac{\pi}{2}$

$=\frac{3\pi}{4}$

15.

$$\begin{aligned} & \sum_{n=1}^{\infty} \frac{1}{n(n+1)} \\ &= \lim_{k \rightarrow \infty} \sum_{n=1}^k \left(\frac{1}{n} - \frac{1}{n+1} \right) \\ &= \lim_{k \rightarrow \infty} \left(1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \dots + \frac{1}{k} - \frac{1}{k+1} \right) \\ &= \lim_{k \rightarrow \infty} \left(1 + \frac{1}{k+1} \right) \\ &= 1 \end{aligned}$$

16.

$$\begin{aligned} & \sum_{n=1}^{\infty} \tan^{-1} \left(\frac{\frac{1}{n(n+1)}}{1 + \frac{1}{n(n+1)}} \right) \\ &= \lim_{k \rightarrow \infty} \sum_{n=1}^k \tan^{-1} \left(\frac{\frac{1}{n} - \frac{1}{n+1}}{1 + \frac{1}{n(n+1)}} \right) \\ &= \lim_{k \rightarrow \infty} \sum_{n=1}^k \left(\tan^{-1} \left(\frac{1}{n} \right) - \tan^{-1} \left(\frac{1}{n+1} \right) \right) \\ &= \lim_{k \rightarrow \infty} \tan^{-1}(1) - \tan^{-1} \left(\frac{1}{2} \right) + \tan^{-1} \left(\frac{1}{2} \right) - \tan^{-1} \left(\frac{1}{3} \right) \\ & \quad + \dots + \tan^{-1} \left(\frac{1}{k} \right) - \tan^{-1} \left(\frac{1}{k+1} \right) \\ &= \lim_{k \rightarrow \infty} \tan^{-1}(1) - \tan^{-1} \left(\frac{1}{k+1} \right) \\ &= \frac{\pi}{4} \end{aligned}$$

참고. $\tan^{-1} m_1 - \tan^{-1} m_2 = \tan^{-1} \left(\frac{m_1 - m_2}{1 + m_1 m_2} \right) \quad (m_1, m_2 > 0, m_1 > m_2)$

17.

$$\begin{aligned} & \sum_{n=1}^{\infty} \frac{1}{2^n} \tan \frac{1}{2^{n-1}} \\ &= \lim_{k \rightarrow \infty} \sum_{n=1}^k \frac{1}{2^n} \tan \frac{1}{2^{n-1}} \\ &= \lim_{k \rightarrow \infty} \sum_{n=1}^k \frac{1}{2^n} \left(\cot \left(\frac{1}{2^{n-1}} \right) - 2 \cot \left(\frac{1}{2^{n-2}} \right) \right) \\ &= \lim_{k \rightarrow \infty} \sum_{n=1}^k \left(\frac{1}{2^n} \cot \left(\frac{1}{2^{n-1}} \right) - \frac{1}{2^{n-1}} \cot \left(\frac{1}{2^{n-2}} \right) \right) \\ &= \lim_{k \rightarrow \infty} \frac{1}{2} \cot \left(\frac{1}{2^0} \right) - \frac{1}{2^0} \cot \left(\frac{1}{2^{-1}} \right) + \frac{1}{2^2} \cot \left(\frac{1}{2} \right) - \frac{1}{2} \cot \left(\frac{1}{2^0} \right) \\ & \quad + \dots + \frac{1}{2^k} \cot \left(\frac{1}{2^{k-1}} \right) - \frac{1}{2^{k-1}} \cot \left(\frac{1}{2^{k-2}} \right) \\ &= \lim_{k \rightarrow \infty} -\frac{1}{2^0} \cot \left(\frac{1}{2^{-1}} \right) + \frac{1}{2^2} \cot \left(\frac{1}{2^{k-1}} \right) \\ &= -\cot 2 + \frac{1}{2} \lim_{n \rightarrow \infty} \frac{\frac{1}{2^{k-1}}}{\tan \left(\frac{1}{2^{k-1}} \right)} = -\cot 2 + \frac{1}{2} \end{aligned}$$

참고. $\tan \left(\frac{x}{2} \right) = \cot \left(\frac{x}{2} \right) - 2 \cot x$

18.

$\sin 3x = 3 \sin x - 4 \sin^3 x$ 이므로

$\sin^3 x = \frac{3}{4} \sin x - \frac{1}{4} \sin 3x$ 이다.

$$\begin{aligned} & \sum_{n=0}^{\infty} 3^n \sin^3 \left(\frac{\pi}{3^{n+1}} \right) \\ &= \lim_{k \rightarrow \infty} \sum_{n=0}^k 3^n \left(\frac{3}{4} \sin \left(\frac{\pi}{3^{n+1}} \right) - \frac{1}{4} \sin \left(\frac{\pi}{3^n} \right) \right) \\ &= \frac{1}{4} \lim_{k \rightarrow \infty} \sum_{n=0}^k \left(3^{n+1} \sin \left(\frac{\pi}{3^{n+1}} \right) - 3^n \sin \left(\frac{\pi}{3^n} \right) \right) \\ &= \frac{1}{4} \lim_{k \rightarrow \infty} 3 \sin \left(\frac{\pi}{3} \right) - 3^0 \sin \left(\frac{\pi}{3^0} \right) + 3^2 \sin \left(\frac{\pi}{3^2} \right) - 3 \sin \left(\frac{\pi}{3} \right) \end{aligned}$$

$$+ \dots + 3^{k+1} \sin \left(\frac{\pi}{3^{k+1}} \right) - 3^k \sin \left(\frac{\pi}{3^k} \right)$$

$$= \frac{1}{4} \lim_{k \rightarrow \infty} \left(-3^0 \sin \left(\frac{\pi}{3^0} \right) + 3^{k+1} \sin \left(\frac{\pi}{3^{k+1}} \right) \right)$$

$$= \frac{1}{4} \lim_{k \rightarrow \infty} \left(\frac{\sin \left(\frac{\pi}{3^{k+1}} \right)}{\frac{\pi}{3^{k+1}} \cdot \frac{1}{\pi}} \right)$$

$$= \frac{\pi}{4}$$

참고. $\sum_{n=0}^{\infty} 3^n \sin^3 \left(\frac{x}{3^{n+1}} \right) = \frac{1}{4} (x - \sin x)$

19.

$\sin \alpha \sin \beta = -\frac{1}{2} (\cos(\alpha + \beta) - \cos(\alpha - \beta))$ 이므로

$$\sin \left(\frac{x}{2} \right) (\sin x + \sin 2x + \dots + \sin 11x)$$

$$= \sin \left(\frac{x}{2} \right) \sin x + \sin \left(\frac{x}{2} \right) \sin 2x + \dots + \sin \left(\frac{x}{2} \right) \sin 11x$$

$$= -\frac{1}{2} \left(\cos \left(\frac{3}{2} x \right) - \cos \left(\frac{1}{2} x \right) + \cos \left(\frac{5}{2} x \right) - \cos \left(\frac{3}{2} x \right) + \dots + \cos \left(\frac{23}{2} x \right) - \cos \left(\frac{21}{2} x \right) \right)$$

$$= -\frac{1}{2} \left(-\cos \left(\frac{1}{2} x \right) + \cos \left(\frac{23}{2} x \right) \right)$$

$$\therefore \sin x + \sin 2x + \dots + \sin 11x$$

$$= \frac{-\frac{1}{2} \left(-\cos \left(\frac{x}{2} \right) + \cos \left(\frac{23}{2} x \right) \right)}{\sin \left(\frac{x}{2} \right)}$$

$$\Rightarrow \sin \left(\frac{\pi}{12} \right) + \sin \left(\frac{2\pi}{12} \right) + \dots + \sin \left(\frac{11\pi}{12} \right)$$

$$= \frac{-\frac{1}{2} \left(-\cos \left(\frac{\pi}{24} \right) + \cos \left(\frac{23\pi}{24} \right) \right)}{\sin \left(\frac{\pi}{24} \right)}$$

$$= \frac{\sin \left(\frac{\pi}{24} \right)}{-\frac{1}{2} \left(-\cos \left(\frac{\pi}{24} \right) + \cos \left(\pi - \frac{\pi}{24} \right) \right)}$$

$$= \frac{-\frac{1}{2} \left(-\cos \left(\frac{\pi}{24} \right) - \cos \left(\frac{\pi}{24} \right) \right)}{\sin \left(\frac{\pi}{24} \right)}$$

$$= \frac{\cos \left(\frac{\pi}{24} \right)}{\sin \left(\frac{\pi}{24} \right)} = \cot \left(\frac{\pi}{24} \right)$$

20. $f(x) = {}_{10}C_1 x^1 + {}_{10}C_2 x^2 + \dots + {}_{10}C_{10} x^{10}$

$$= (1+x)^{10} - 1 \quad (\because \text{이항정리})$$

$$\therefore f \left(\frac{1}{2} \right) = \left(\frac{3}{2} \right)^{10} - 1$$